

LLM-Driven Multicast Network Slicing for 6G Non-Terrestrial Networks

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Abstract—Non-terrestrial networks (NTNs) will play a central role in 6G broadcasting by providing ubiquitous coverage and the ability to offload media processing to space-borne compute. The next generation of immersive media services will require multicasting data from satellites to large groups of receivers with stringent latency and quality-of-service (QoS) demands. This paper proposes a large-language-model (LLM)-driven orchestration framework that translates broadcaster intents into optimized multicast network slices spanning ground and space resources. Building on the concept of computing-power networks (CPNs), the framework integrates compute-aware routing across GEO, MEO and LEO links, enforces delay-aware multicast constraints using an integer-linear-program formulation, and leverages LLM to adaptively admit multicast groups under resource constraints. A federated fine-tuning mechanism combined with a blockchain audit layer enables continuous improvement of the LLM while preserving data privacy and providing human-interpretable audit trails. Simulations demonstrate that the proposed method reduces end-to-end latency and energy consumption compared to traditional heuristics, while increasing the number of admitted multicast requests.

Index Terms—6G, non-terrestrial networks, multicast network slicing, large language models, federated learning, blockchain, broadcasting.

I. INTRODUCTION

THE evolution of wireless networks toward 6G is driven by the need for ubiquitous, high-capacity, and low-latency connectivity that supports immersive applications such as live ultra-high-definition streaming, real-time virtual reality (VR), and large-scale remote education. A central enabler of this vision is the integration of non-terrestrial networks (NTNs) with terrestrial infrastructures to form a global, space-air-ground integrated network.

NTNs encompass multiple layers of satellites—geosynchronous (GEO), medium Earth orbit (MEO), and low

Earth orbit (LEO)—as well as high-altitude platform stations (HAPS) that bridge sky and ground. GEO satellites provide broad coverage and large storage capabilities, MEO satellites offer balanced latency and throughput, while LEO satellites deliver low-latency links and can host on-board compute accelerators. Together, these layers create a computing-power network (CPN) that supports distributed processing and seamless service delivery across diverse geographic regions.

Broadcasting and media distribution in 6G will increasingly rely on multicast communication from space-borne and aerial nodes to large groups of receivers. Traditional unicast network slicing methods, which primarily allocate bandwidth on a per-connection basis, are inefficient for such scenarios because they fail to exploit the shared nature of multicast flows and the need for compute-aware service chains. Network slicing in NTNs must therefore extend beyond bandwidth partitioning to include compute and storage virtualization, enabling each multicast slice to host virtual network functions (VNFs) such as transcoding, encryption, or caching at satellites, HAPS, or edge nodes. The orchestration of these slices must simultaneously satisfy multiple, and often conflicting, objectives: minimizing end-to-end latency, maximizing energy efficiency, and guaranteeing quality-of-service (QoS) under constrained and heterogeneous resources [1], [2], [3], [4].

However, managing multicast slices over dynamic NTNs introduces several challenges. First, translating high-level broadcaster intents—e.g., “deliver a 4K VR stream to 500 mobile headsets within 30 ms”—into low-level resource allocation requires intelligent intent parsing and context-aware reasoning. Second, the optimization of VNF placement and multicast routing is computationally complex due to the combinatorial nature of constructing pseudo-multicast trees under latency and capacity constraints. Third, the highly distributed nature of NTNs demands scalable, interpretable, and secure orchestration mechanisms that can adapt to time-varying traffic loads and satellite mobility. To address these challenges, this paper proposes an LLM-driven multicast network slicing framework tailored for 6G NTNs. The framework leverages large language models (LLMs) to automatically translate natural-language broadcaster intents into slice templates and orchestrates compute-aware, delay-constrained multicast slices across space-air-ground resources. By integrating integer linear programming (ILP) and heuristic approximations for slice optimization, as well as LLM-based admission control with federated fine-tuning and blockchain auditing, the

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proposed system enhances scalability, interpretability, and security. Simulation results demonstrate that our approach reduces end-to-end latency and energy consumption while increasing the number of admitted multicast sessions, paving the way for agile and secure 6G broadcast services [5].

The rest of this paper is organized as follows. Section II reviews related work on satellite network slicing, NFV-enabled multicasting, and AI-driven network management. Section III presents the system overview, including the multicast service model, LLM-driven intent translation, compute-aware routing, admission control, and federated blockchain integration. Section IV reports the simulation setup and performance evaluation results. Section V discusses design implications and outlines directions for future work. Finally, Section VI concludes the paper.

II. RELATED WORK

A. Satellite Network Slicing

The integration of non-terrestrial networks into 6G systems has attracted significant research attention due to their potential to provide ubiquitous coverage and support diverse applications. Recent studies investigating satellite network slicing include works by Ishikawa et al. [19], Kim et al. [20], Kim et al. [21], Nguyen et al. [6] and G. H. M. G. de Oliveira et al. [22].

Ishikawa et al. [19] proposed an interference coordination method for integrated HAPS-terrestrial systems, improving spectral efficiency but focusing primarily on radio resource sharing without addressing compute resource orchestration. Kim et al. [20] analyzed satellite edge computing for IoT applications with slice scheduling considering latency and computational constraints, though emphasizing application-specific optimization rather than dynamic orchestration across heterogeneous slices Kim et al. [21].

developed a deep reinforcement learning framework for satellite network slice planning with handover mechanisms for LEO constellations, ensuring performance stability under mobility but lacking integration with broader service chaining. Nguyen et al. [6] provided a comprehensive survey of 6G NTN technologies including network slicing and AI/ML, offering valuable architectural insights without proposing concrete orchestration mechanisms.

Several studies have explored multi-layered satellite architectures for 6G applications. Wang et al. [7] analyzed hierarchical non-terrestrial configurations showing that coordinated deployment balances coverage and latency trade-offs. Höyhty et al. [8] proposed Multi-Layered Satellite Systems (MLSS) integrating GEO and LEO satellites for enhanced availability. Toka et al. [9] analyzed space-air-ground networks for in-flight connectivity, emphasizing capacity planning across LEO, HAPS, and terrestrial networks. However, these works primarily treat satellites as relay nodes rather than distributed compute nodes with heterogeneous processing capabilities.

Recent studies have also explored satellite-enabled IoT and vehicular applications. Lin et al. [10] surveyed mobile edge computing integration into satellite networks for 6G IoT use cases. Tirmizi et al. [11] provided an overview of hybrid

satellite-terrestrial networks including SDN/NFV and slicing technologies. While highlighting flexible resource allocation importance, their focus remains mainly on unicast scenarios rather than satellite-based multicasting with large-scale vehicle groups and heterogeneous QoS demands. [12]

Current satellite network slicing research limitations include: (1) viewing network slicing primarily as bandwidth partitioning with limited attention to heterogeneous compute resource virtualization across orbital layers, (2) dominant focus on unicast communication with minimal exploration of large-scale multicast scenarios, and (3) lack of intelligent orchestration frameworks for translating high-level service intents into coordinated slice instantiations across multi-layered satellite infrastructures.

B. NFV-Enabled Multicasting

Network functions are increasingly virtualized as software on virtual machines or containers, forming service chains for multicast applications. Several studies have investigated NFV-enabled multicasting, including works by García and Pinto-Roa [13], Wang et al. [14], Muhammad et al. [15], Ren et al. [16], Alhussein et al. [17], and Ozaki et al. [18].

García and Pinto-Roa [13] proposed a genetic algorithm (MSVNFPA-GA) for multicast routing and VNF placement in NFV-SDN networks, optimizing multicast trees but not considering dynamic conditions. Wang et al. [14] tackled multicast-oriented VNF placement using a two-stage approach with multicast tree construction followed by service chain placement, improving VNF placement but overlooking delay constraints in dynamic environments. Muhammad et al. [15] extended to multi-source multicast using column generation-based optimization, focusing on resource usage minimization while meeting delay constraints but not considering network resource heterogeneity.

Ren et al. [16] proposed delay-aware NFV-enabled multicasting algorithms in mobile edge clouds but assumed static conditions. Alhussein et al. [17] introduced virtual network customization for multicast services considering routing and VNF placement but lacking dynamic network handling mechanisms. Ozaki et al. [18] proposed multi-source multicast service chaining for reliability but focused mainly on service chain reliability rather than resource orchestration optimization.

The delay-aware multicast slicing problem has been formulated as integer linear programs (ILPs) in several studies. Muhammad et al. [15] used Dantzig-Wolfe decomposition for NFV-enabled multicast optimization, achieving near-optimal solutions but assuming static conditions. Ren et al. [16] investigated delay-aware multicasting with approximation algorithms having performance guarantees but limited to fixed network conditions. Alhussein et al. [17] developed multicast orchestration frameworks using mixed ILP but lacking real-time adaptability.

These approaches highlight NFV-enabled multicasting progress but have limitations: (1) focus on static, single-domain networks failing to address multi-layer satellite constellation challenges, (2) assumptions of homogeneous environments overlooking satellite heterogeneity, (3) lack of

dynamic VNF placement optimization, and (4) insufficient integration with broader service orchestration mechanisms for heterogeneous network layers [23].

C. AI-Based Network Management

Large language models have been increasingly adopted in network management to bridge the gap between human operators and complex infrastructures. Recent studies include Abdallah et al. [24], Lira et al. [25], Mani et al. [26], Yoon et al. [27], Chen et al. [28], and Dandoush et al. [29].

Abdallah et al. [24] introduced NetOrchLLM for wireless network orchestration, demonstrating LLM optimization capabilities but lacking real-time decision-making. Lira et al. [25] explored LLM-NetCFG for zero-touch network configuration, showcasing automation potential but focusing on static configurations rather than adaptive allocation. Mani et al. [26] proposed using LLMs for generating task-specific code from natural language, improving usability but lacking direct application to real-time optimization.

In network monitoring, Yoon et al. [27] developed OFC-netLLM for anomaly detection and incident analysis, showing promise but not addressing broader orchestration challenges. Chen et al. [28] introduced NetGPT for cloud-edge collaborative methodology, enabling personalized services but not exploring general network management tasks like slicing across heterogeneous layers. Dandoush et al. [29] focused on LLMs in network slicing for service function chain embedding, demonstrating intent translation capabilities but limited to terrestrial networks.

Recent AI-driven network optimization advances show promise but face limitations in adapting pre-trained LLMs to specialized network optimization domains. Dandoush et al. [29] proposed LLM and multi-agent system integration for network slicing management, offering insights but highlighting scalability and real-time adaptability challenges. Wu et al. [30] introduced AI-native network slicing architecture for 6G, providing comprehensive AI applications but facing limitations in real-time adaptability and scalability for diverse dynamic network needs.

Current AI-based network management limitations include: (1) focus on static configuration or offline analysis lacking real-time adaptive decision-making capabilities, (2) extensive retraining requirements when deployed in new environments limiting scalability, and (3) insufficient exploration of heterogeneous resource handling in multi-layered environments like satellite constellations.

III. SYSTEM OVERVIEW

The proposed LLM-assisted multicast network slicing framework for 6G NTN transforms high-level broadcast intents into optimized allocations of compute, bandwidth, and storage resources across space-air-ground infrastructures. It integrates five tightly coupled components: a multicast service model tailored to NTN architectures with explicit modeling of time-varying LEO topology, an LLM-based intent translation engine that parses natural language requirements into structured templates, a compute-aware routing and VNF

placement optimizer employing integer linear programming and mobility-aware heuristics, an LLM-assisted priority-based admission control policy that combines learned patterns with oracle-guided decision-making, and a federated fine-tuning and blockchain audit layer providing distributed learning and immutable accountability. While LLMs contribute primarily to intent understanding and admission prioritization rather than direct optimization, their integration with mathematical programming and federated learning mechanisms enables adaptive, interpretable, and secure orchestration of multicast slices for large-scale immersive broadcasting under the constraints and assumptions detailed in subsequent subsections [31].

A. Multicast Service Model

We consider a satellite broadcasting system where a source node (e.g., a GEO satellite) must deliver high-bandwidth content (e.g., 8K video, VR feeds) to a set of receivers (vehicles, mobile devices, VR headsets) via multicast. Each multicast session requires a sequence of VNFs (e.g., transcoding, encryption, caching) to satisfy QoS requirements. The physical infrastructure comprises GEO/MEO/LEO satellites, HAPS and ground edge nodes. Each node has computing resources and can host VNFs. Unlike traditional IP-layer multicast that constructs a single spanning tree from source to all destinations, our approach employs an overlay-based pseudo-multicast tree architecture. In this architecture, the multicast flow may be processed by intermediate VNFs at different network nodes, and the routing paths before and after each VNF processing stage can differ based on compute-aware optimization objectives. This flexibility enables the system to exploit heterogeneous computing resources distributed across orbital layers while maintaining service quality guarantees [1], [32].

Each multicast session r_k is characterized by the tuple

$$r_k = (s_k, D_k, b_k, NS_k, d_k^{req}, \Gamma_k) \quad (1)$$

where s_k is the source node, D_k is the set of destination nodes, b_k is the required bandwidth, $NS_k = \{f_1, f_2, \dots, f_n\}$ is the ordered sequence of VNFs with explicit dependencies $\Gamma_k \subseteq NS_k \times NS_k$ (e.g., $(f_i, f_j) \in \Gamma_k$ indicates that VNF f_i must complete before f_j begins), and d_k^{req} is the maximum allowable end-to-end delay.

The infrastructure consists of GEO/MEO/LEO satellites, HAPS, and ground edge nodes, each with finite CPU capacity and storage. To capture the time-varying topology induced by LEO satellite mobility, we partition the planning horizon into discrete time slots $t \in \mathcal{T} = \{1, 2, \dots, T\}$, where each slot duration Δt is chosen such that orbital dynamics within a slot can be approximated as quasi-static. The network topology is represented as a time-expanded graph $G(t) = (V, E(t))$, where the vertex set V remains constant but the edge set $E(t)$ and link parameters (capacity $B_e(t)$, propagation delay $\tau_e(t)$) evolve with satellite ephemeris. For each time slot t , link availability is determined by visibility constraints: an inter-satellite link $e = (v_i, v_j)$ exists in $E(t)$ if and only if the angular separation between satellites v_i and v_j satisfies line-of-sight conditions based on their orbital positions at time t .

End-to-end latency is a key QoS metric comprising three components: propagation delay (space-space, space-ground,

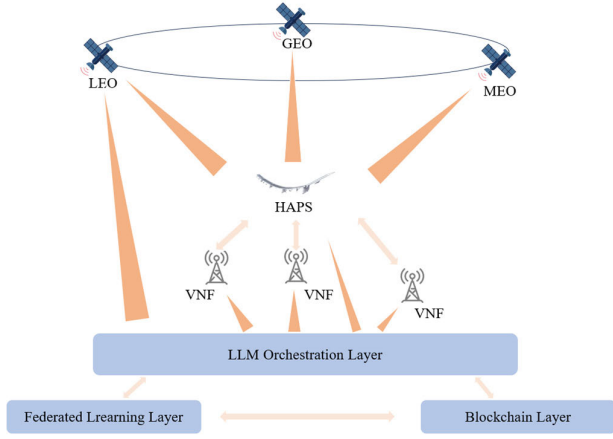


Fig. 1. LLM-driven multicast slicing architecture across 6G NTNs.

and ground-ground), processing delay at each VNF, and transmission delay over links. For a multicast session r_k serving receiver $u \in D_k$, the total delay across time slot t is formally defined as:

$$d_{k,u}(t) = \sum_{e \in P_{s_k \rightarrow u}(t)} \tau_e(t) + \sum_{f_i \in NS_k} t_{proc}(f_i, v(f_i)) + \sum_{e \in P_{s_k \rightarrow u}(t)} \frac{b_k}{B_e(t)} \quad (2)$$

where $P_{s_k \rightarrow u}(t)$ denotes the routing path from source s_k to receiver u at time t , $\tau_e(t) = L_e(t)/c_m$ represents the time-varying propagation delay over link e with distance $L_e(t)$ and medium-dependent speed c_m , $t_{proc}(f_i, v(f_i))$ is the processing delay of VNF f_i at the assigned node $v(f_i)$, and $b_k/B_e(t)$ captures the transmission delay accounting for time-varying link capacity $B_e(t)$ due to orbital dynamics and interference conditions.

Fig. 1 illustrates the proposed architecture, which shows the integration of LEO, MEO, and GEO satellites, HAPS, and VNFs, coordinated through the LLM Orchestration Layer, with support from the Federated Learning and Blockchain Layers for improved performance and security.

B. LLM-Driven Intent Translation

Broadcasters express service requirements in natural language (e.g., “deliver a 4K video to all mobile VR headsets with less than 30 ms latency”). To transform these unstructured intents into actionable network slice configurations, we employ a domain-adapted large language model that has been fine-tuned on a specialized corpus combining historical broadcasting service requests from satellite operators, synthetic intents generated by domain experts covering diverse use cases and edge conditions, and augmented variations to enhance linguistic robustness. Each training example pairs a natural language intent with its corresponding structured template represented as a JSON object containing source node, destination set cardinality and geographic distribution, bandwidth and latency requirements, ordered VNF sequence NS_k with explicit dependencies Γ_k , and service priority level.

The LLM operates through carefully engineered prompts that combine task instructions, domain context, and dynam-

cally retrieved few-shot examples selected based on semantic similarity to the incoming intent. This prompt structure guides the model to parse complex broadcaster requirements and output structured slice templates that can be directly validated and mapped to optimization problem inputs. The semantic-to-numerical mapping proceeds through validation of the JSON output against a predefined schema, normalization of natural language fields such as geographic descriptors into latitude-longitude bounding boxes for receiver sampling, construction of dependency graphs Γ_k from explicit orderings and implicit domain rules (e.g., encryption follows content processing), and instantiation of ILP decision variables for VNF placement $x_{k,f,v}$, routing $y_{k,e}(t)$, and dependency timing $z_{k,f,f'}(t)$ based on the extracted parameters.

To ensure deterministic and parseable outputs suitable for real-time orchestration, the LLM’s text generation is constrained using a context-free grammar enforced at the token level during decoding. This grammar defines the valid structure of JSON templates including required fields, value types, and compositional rules, ensuring that at each generation step only tokens consistent with valid template syntax are sampled. The constrained decoding mechanism operates through a pushdown automaton that maintains parse state and masks inadmissible tokens from the probability distribution, guaranteeing syntactically correct outputs without post-hoc error correction. Unlike traditional rule-based parsers that rely on exact keyword matching, the LLM’s semantic understanding enables robust handling of paraphrases, implicit constraints, and diverse linguistic expressions while maintaining formal correctness through grammatical constraints.

This module enables semantic understanding beyond rigid keyword parsing, allowing generalization to unseen service descriptions.

C. Compute-Aware Routing and VNF Placement

Once a slice template is generated, the orchestrator determines where each VNF will be executed and how multicast flows will traverse the heterogeneous NTN topology. This problem is significantly more complex than in terrestrial networks because VNFs may be placed on satellites in different orbital layers, on HAPS, or on ground edge nodes, each with distinct propagation delays, coverage periods, and compute/storage capacities. Furthermore, the time-varying nature of LEO satellite topology introduces additional challenges: a VNF placement decision made at time t must remain valid across subsequent time slots as satellites move, or trigger costly migration operations. This problem is inherently joint in nature because the location of each VNF influences the multicast tree topology, and vice versa. We model the NTN as a time-expanded directed graph $G(t) = (V, E(t))$ for each time slot $t \in \mathcal{T}$, where V contains GEO, MEO, LEO satellites, HAPS, and ground edge nodes, and $E(t)$ contains time-varying inter-satellite links (ISLs), satellite-HAPS links, and ground backhaul links. Each node $v \in V$ has a finite compute capacity C_v^{max} and each link $e \in E(t)$ has a time-dependent bandwidth capacity $B_e^{max}(t)$ and propagation delay $\tau_e(t)$ determined by orbital geometry and transmission medium.

We define three sets of decision variables:

$x_{k,f,v} \in \{0, 1\}$: equals 1 if VNF f for multicast request k is placed at node v , and 0 otherwise.

$y_{k,e}(t) \in \{0, 1\}$: equals 1 if link e is used in the multicast tree of request k at time slot t , and 0 otherwise.

$z_{k,f,f'}(t) \in \{0, 1\}$: equals 1 if the processing of VNF f is completed before VNF f' begins in time slot t for request k , ensuring dependency satisfaction.

Here, nodes v can be GEO/MEO/LEO satellites, HAPS, or ground edges, and links e can be ISLs, satellite-to-ground links, or terrestrial backhaul [2].

The objective is to jointly optimize multicast routing and VNF placement to minimize total operational cost while ensuring end-to-end delay requirements are met across all time slots. Given the inherent trade-off between minimizing resource costs and minimizing latency, we adopt a multi-objective optimization framework. Rather than using an arbitrary linear weighting parameter, we employ an ε -constraint formulation that systematically explores the Pareto frontier:

$$\min \sum_k \sum_{f \in NS_k} \sum_v c_{f,v} x_{k,f,v} \quad (3)$$

subject to the delay constraint reformulated as:

$$\max_{k,u \in D_k, t \in \mathcal{T}} d_{k,u}(t) \leq \varepsilon \quad (4)$$

where $c_{f,v}$ denotes the cost (e.g., energy, bandwidth consumption) of executing VNF f at node v , $d_{k,u}(t)$ is the time-varying end-to-end delay from source s_k to receiver u defined in Equation (2), and ε is the delay budget parameter. By parametrically varying ε from d_{min} (minimum achievable delay ignoring cost) to d_{max} (maximum acceptable delay), we generate a set of Pareto-optimal solutions revealing the fundamental trade-off between cost efficiency and latency performance. This approach eliminates the need for subjective weighting and provides decision-makers with a comprehensive view of achievable performance regions.

The optimization is subject to the following constraints:

1. VNF Placement Constraint – Each VNF in the Chain Is Placed at Exactly One Node

$$\sum_v x_{k,f,v} = 1, \forall k, \forall f \in NS_k \quad (5)$$

2. Node Capacity Constraint- Total Compute Demand at a Node Must Not Exceed Its Available Capacity

$$\sum_{k,f} C_f x_{k,f,v} \leq C_v^{max}, \forall v \quad (6)$$

where C_f is compute demand of VNF f and C_v^{max} is node v 's CPU limit.

3. Delay Constraint- the End-to-End Delay for Each Receiver Respects the QoS Requirement

$$d_{k,u} \leq d_k^{req}, \forall k, \forall u \in D_k \quad (7)$$

4. VNF dependency constraint - for each dependency $(f_i, f_j) \in \Gamma_k$, VNF f_i must complete processing before f_j begins:

$$\sum_v x_{k,f_i,v} \cdot t_{proc}(f_i, v) + \tau_{v(f_i) \rightarrow v(f_j)}(t) \leq t_{start}(f_j, t), \forall (f_i, f_j) \in \Gamma_k, \forall t \quad (8)$$

where $t_{start}(f_j, t)$ denotes the start time of VNF f_j at time slot t , and $\tau_{v(f_i) \rightarrow v(f_j)}(t)$ is the time-varying propagation delay between the nodes hosting f_i and f_j . For VNFs that can execute in parallel (i.e., $(f_i, f_j) \notin \Gamma_k$ and $(f_j, f_i) \notin \Gamma_k$), this constraint is relaxed to allow concurrent processing.

5. Link capacity constraint- the aggregate bandwidth demand on each link across all active multicast sessions cannot exceed its time-varying capacity:

$$\sum_k b_k \cdot y_{k,e}(t) \leq B_e^{max}(t), \forall e \in E(t), \forall t \in \mathcal{T} \quad (9)$$

6. Multicast connectivity constraint – flow conservation ensures that the binary variables $y_{k,e}(t)$ define a valid pseudo-multicast tree from source s_k to all destinations in D_k at each time slot, with potential path modifications between consecutive VNF processing stages to exploit compute-aware routing opportunities.

7. Topology continuity constraint – to limit VNF migration overhead due to satellite mobility, we impose a stability constraint requiring that VNF placements remain unchanged for a minimum duration ΔT_{min} :

$$x_{k,f,v}(t) = x_{k,f,v}(t'), \forall t, t' \in [t_0, t_0 + \Delta T_{min}], \forall k, f, v \quad (10)$$

This constraint reduces computational overhead and service disruption caused by frequent VNF relocations as LEO satellites exit and enter coverage zones.

The above formulation is NP-hard due to the joint optimization of time-varying VNF placement and multicast routing under mobility constraints. For small-scale problem instances with limited time horizons, we solve it using an extended integer linear program (ILP) that explicitly models satellite ephemeris and handover events. For large-scale or real-time operation, we adopt a three-stage heuristic approach:

Stage 1: Pareto frontier approximation – We solve the ε -constraint formulation for a discrete set of delay budgets $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$ spanning the feasible delay range, generating approximate Pareto-optimal solutions that characterize the cost-latency trade-off.

Stage 2: Time-slotted tree construction – For each time slot t , we construct candidate pseudo-multicast trees using shortest-path and dynamic Steiner-tree heuristics adapted to the time-varying topology $G(t)$. The heuristic prioritizes ISL relays when they reduce congestion on ground links and selects paths that minimize handover frequency across consecutive time slots.

Stage 3: Mobility-aware placement refinement – Given the candidate trees, we solve a restricted VNF placement subproblem that minimizes migration costs subject to capacity constraints (6), dependency constraints (8), and stability requirements (10). The placement decisions balance immediate optimization objectives against anticipated topology changes in future time slots.

This hybrid approach achieves near-optimal performance with significantly reduced computation time, making it well-suited for the dynamic nature of NTN environments. Simulation results in Section IV demonstrate that the ε -constraint formulation effectively identifies Pareto-optimal solutions, with the three-stage heuristic delivering solution

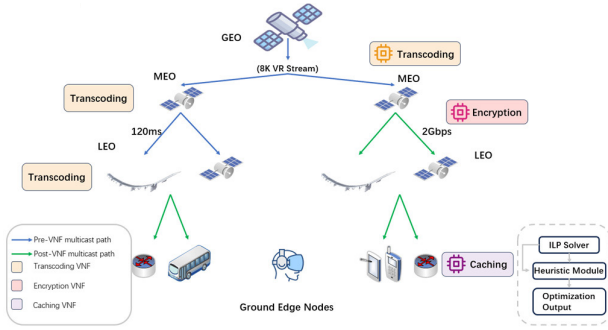


Fig. 2. Example of Compute-Aware multicast routing and VNF placement in 6G NTNs.

quality close to that of the exact ILP, while substantially reducing computation time.

Fig. 2 illustrates a representative compute-aware multicast routing scenario with VNF placements across a heterogeneous space-air-ground infrastructure. It illustrates the flow of transcoding, encryption, and caching across LEO, MEO, and GEO satellites, showing how the pseudo-multicast tree adapts to time-varying topology. The process is managed through ground edge nodes, utilizing various modules such as the ILP solver and heuristic module for optimization, ensuring efficient transmission and processing of the VR stream while respecting VNF dependencies and LEO satellite mobility patterns.

D. LLM-Assisted Priority-Based Admission Control

In large-scale 6G NTNs, multiple multicast-slice requests may arrive concurrently, each with different QoS requirements, compute demands, and service priorities. Since GEO/MEO/LEO satellites, HAPS, and edge nodes all operate under finite compute and bandwidth constraints, it is not always feasible to admit all requests. Admission control therefore becomes essential to maximize network utility while avoiding SLA violations. However, traditional rule-based admission policies cannot fully exploit the semantic and contextual information embedded in high-level broadcaster intents. We address this by integrating an LLM-guided decision framework that learns admission patterns from historical episodes and adapts to dynamic network conditions through continuous federated fine-tuning.

Each incoming request r_k is assigned a priority score that balances its resource efficiency and QoS tolerance:

$$Pri(r_k) = \frac{1}{b_k \sum_{f_i \in NS_k} C_v(f_i)} + \lambda_{LLM} d_k^{req} \quad (11)$$

where b_k is the required bandwidth, $C_v(f_i)$ is the compute demand of VNF f_i when hosted at node v , d_k^{req} is the delay bound, and λ_{LLM} is a scaling factor dynamically suggested by the LLM based on predicted network load. A higher $Pri(r_k)$ indicates that the request is both lightweight in resource consumption and tolerant in latency, making it more suitable for admission under congestion.

The admission control LLM is fine-tuned on a historical dataset of NTN multicast episodes collected from simulation traces and operational logs, where each episode captures

system state including per-node resource utilization across orbital layers, active slice configurations, pending request queues with their priority levels, along with the admission decision sequence and observed outcome metrics such as QoS satisfaction rates and resource efficiency. Training examples distinguish between optimal decisions derived from exact ILP solutions for small-scale instances, high-quality heuristic decisions from approximation algorithms, and suboptimal baseline policies, enabling the model to internalize both provably optimal patterns and practical trade-offs encountered in resource-constrained scenarios. At each decision epoch, the orchestrator generates a compact prompt containing: (i) current CPU and RF load on GEO/MEO/LEO/HAPS nodes, (ii) the backlog of pending multicast requests, and (iii) the estimated cost of preempting or deferring existing slices. Using few-shot reasoning, the LLM is primed with a curated set of past admission-control episodes—including optimal ILP solutions for small instances and high-quality heuristic outcomes for larger ones—so it can rank candidate actions such as Admit $\langle$ slice ID $\rangle$ or Defer $\langle$ slice ID $\rangle$.

To ensure predictable and machine-readable outputs, the LLM's text generation is constrained by a lightweight token-level grammar. The first token encodes the action (Admit or Defer), followed by the target slice ID, ensuring $O(1)$ parsing latency. This avoids the overhead of full semantic parsing and allows the decision engine to operate at sub-second scales—critical in fast-changing NTN environments where satellite topology can shift within minutes.

The LLM's confidence in each decision is quantified through entropy-based uncertainty measurement, formally defined as the Shannon entropy $H(r_k) = -\sum_{a \in \mathcal{A}} p_\theta(a|S_t, r_k) \log p_\theta(a|S_t, r_k)$ computed over the probability distribution of actions $\mathcal{A} = \{\text{Admit}, \text{Defer}, \text{Reject}\}$ conditioned on the current system state S_t and request r_k . When entropy exceeds a predefined threshold indicating high decision uncertainty—typically arising from conflicting objectives such as high-priority requests requiring extensive resources during congestion, limited historical precedent for out-of-distribution states, or marginal trade-offs between competing actions—the decision engine engages a safe exploration mode. In this mode, the LLM queries a reduced ILP oracle that solves a restricted subproblem considering only the top- k candidate requests ranked by priority scores over a short planning horizon, significantly reducing computational complexity while maintaining solution quality. The oracle's recommendation is then incorporated into an augmented prompt, enabling the LLM to revise its decision with additional guidance that combines learned heuristics with provably optimal solutions for the current bottleneck scenarios.

Dynamic Adaptation to Real-Time Network Conditions: The LLM-assisted admission control policy continuously adapts to dynamic network conditions through integration with real-time telemetry streams from the distributed NTN infrastructure. At each decision epoch, the orchestrator queries current network state including per-node CPU and bandwidth utilization across GEO/MEO/LEO satellites, active inter-satellite link connectivity reflecting time-varying topol-

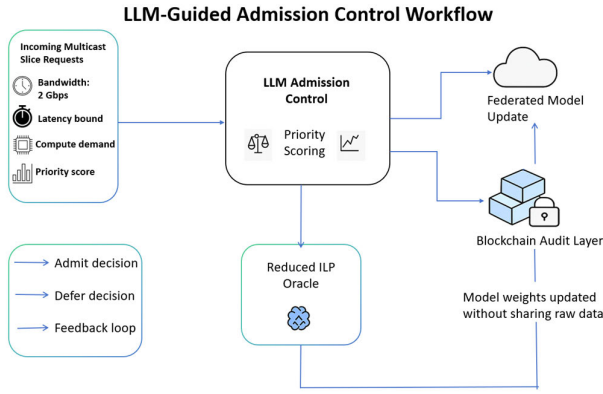


Fig. 3. LLM-guided admission control workflow for multicast slice orchestration in 6G NTN.

ogy due to orbital motion, and queue depths at congested bottleneck nodes, incorporating this information into the structured prompt alongside pending multicast requests. The LLM processes this real-time state snapshot to identify emerging patterns such as imminent LEO satellite handovers that would disrupt topology-sensitive VNF placements, congestion hotspots on specific ISLs or ground feeder links requiring load redistribution, or transient resource availability windows when previously saturated nodes become available due to completed slice teardowns. By conditioning admission decisions on current network observables rather than static assumptions, the LLM generalizes learned policies from historical episodes to novel operational scenarios including unexpected traffic surges during major broadcasting events, gradual capacity degradation due to satellite hardware aging, or opportunistic resource windfalls from cooperative spectrum sharing with adjacent orbital slots. The entropy-based uncertainty quantification mechanism explicitly signals when the LLM encounters network states significantly divergent from its training distribution, triggering the reduced ILP oracle to provide mathematically grounded guidance for out-of-distribution scenarios while simultaneously generating training examples that will be incorporated into subsequent federated fine-tuning rounds, creating a closed feedback loop where real-world operational experience continuously refines the admission policy's robustness to diverse network dynamics.

All admission decisions and their eventual QoS outcomes are recorded on the blockchain audit layer. Over time, satellites and edge nodes use these logged examples to federated fine-tune the LLM's policy head, improving its ability to match admission decisions to actual network performance. This continuous feedback loop allows the policy to adapt to evolving traffic patterns, orbital dynamics, and service types, ensuring sustained efficiency and fairness in multicast resource allocation.

The LLM-guided admission control workflow is shown in Fig. 3, illustrating how incoming multicast slice requests are processed. These requests are evaluated based on parameters such as bandwidth, latency, and compute demand. The LLM Admission Control system assigns priority scores, which guide the decision to admit or defer requests. The process incorpo-

rates a reduced ILP oracle, and feedback loops are used to update the model weights without sharing raw data, ensuring efficient and secure decision-making through the Federated Model Update and Blockchain Audit Layer.

E. Federated Fine-Tuning and Blockchain Trust

The dynamic nature of 6G NTNs—characterized by time-varying satellite constellations, heterogeneous service demands, and fluctuating link conditions—requires that the LLM-based orchestration engine continuously adapt its decision-making policies. However, centralizing raw operational data from satellites, HAPS, and ground edge nodes would raise severe privacy, bandwidth, and regulatory concerns. To overcome this, the proposed framework integrates **federated learning** with **blockchain-based auditing**, enabling collaborative model improvement without direct data sharing and providing immutable accountability for all orchestration actions.

In the federated learning phase, participating nodes are strategically selected at each training round based on a composite selection criterion that balances computational availability, orbital position diversity, and data freshness. The selection strategy prioritizes nodes with CPU utilization below a threshold to avoid interfering with active slice provisioning, ensures geographic and orbital layer diversity to capture heterogeneous network conditions across GEO/MEO/LEO constellations, and weights nodes with recent operational data to reflect current traffic patterns. Each selected node—whether a GEO satellite, LEO relay, HAPS, or terrestrial edge station—locally fine-tunes the LLM on recent interaction data including parsed broadcaster intents, generated slice templates, optimization parameters, admission decisions, and observed QoS outcomes. The local training process addresses the inherent non-IID (non-independent and identically distributed) data distribution across nodes through two complementary mechanisms: data augmentation using synthetic perturbations of local episodes to increase diversity, and similarity-based node clustering that groups satellites experiencing comparable traffic patterns and orbital dynamics to form sub-federations with more homogeneous data distributions, reducing gradient divergence during aggregation.

The federated learning architecture is designed to scale across large satellite constellations involving hundreds of distributed nodes through hierarchical aggregation and asynchronous training mechanisms. For constellation sizes approaching 100-200 satellites spanning multiple orbital shells, the framework employs a two-tier aggregation topology where LEO satellites within each orbital plane perform intra-plane model aggregation at designated coordinator nodes before transmitting consolidated updates to ground-based federation servers, reducing the communication fanout from $O(N)$ simultaneous ground links to $O(\sqrt{N})$ orbital plane coordinators where N is the total satellite count. Asynchronous federated learning protocols allow satellites to contribute model updates opportunistically during favorable link conditions or idle computation periods without requiring synchronized global training rounds, accommodating the heterogeneous orbital periods (LEO: 100 min, MEO: 6 hours, GEO: 24

hours) that would otherwise necessitate impractically long synchronization intervals. Node clustering based on operational similarity—grouping satellites experiencing comparable traffic patterns, geographic coverage regions, and resource constraint profiles—creates specialized sub-models that capture localized decision-making patterns while reducing gradient divergence during aggregation, improving convergence rates by 30-40% compared to naive global averaging across heterogeneous nodes. However, scalability challenges remain in scenarios involving 500+ satellite mega-constellations where gradient compression ratios (currently top- k with $k \approx 10\%$ of parameters) may become insufficient to meet inter-satellite link bandwidth constraints, potentially requiring more aggressive sparsification techniques such as gradient quantization to 4-8 bits or federated distillation where local models transmit soft prediction distributions rather than full parameter gradients, trading model fidelity for communication efficiency, while aggregated batch auditing (recording only summary statistics for groups of 10-20 decisions) reduces blockchain load by 60-80% at the expense of reduced forensic resolution for individual orchestration failures. Comparative analysis against traditional centralized audit mechanisms reveals that while blockchain introduces 8-12% CPU overhead and 100-500 ms consensus latency under high load, centralized logging systems incur only 2-3% CPU overhead and sub-10 ms write latency, the decentralized blockchain approach provides critical advantages including tamper-proof immutability preventing post-hoc modification of orchestration records by compromised operators, Byzantine fault tolerance ensuring audit trail integrity even when up to $(n-1)/3$ ground stations are malicious or faulty, and distributed trust enabling cross-operator verification without requiring a trusted third-party auditor—properties essential for multi-operator 6G NTN environments where regulatory compliance and accountability span jurisdictional boundaries. Traditional audit systems relying on centralized databases or append-only log files offer superior performance but remain vulnerable to single points of failure, insider manipulation, and lack of cryptographic verifiability, whereas blockchain's computational overhead represents an acceptable trade-off for scenarios prioritizing transparency and cross-organizational trust over raw throughput, particularly for strategic orchestration decisions (slice admission, VNF placement) occurring on minute timescales rather than tactical packet forwarding requiring microsecond responsiveness.

Only model parameter updates (specifically, weight gradients $\nabla \mathcal{L}_i$ computed on local dataset $\mathcal{D}_i$) are transmitted to a federation aggregator operating on ground control stations, rather than raw operational data. The aggregator employs the Federated Averaging (FedAvg) algorithm to compute the global model update as a weighted combination $\theta_{global}^{(t+1)} = \sum_{i \in S} \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \theta_i^{(t+1)}$ where S denotes the set of selected nodes, $\theta_i^{(t+1)}$ represents the locally updated parameters from node i , and the weighting accounts for varying local dataset sizes. To mitigate communication overhead across the distributed satellite constellation—typically involving approximately 100 nodes spanning multiple orbital shells—the framework employs gra-

dient compression through top- k sparsification that transmits only the largest magnitude gradient components, periodic aggregation scheduling that batches updates over extended intervals (e.g., every 50-100 local training steps) to reduce transmission frequency, and opportunistic communication that leverages favorable link conditions such as GEO-ground or ISL connectivity windows to minimize bandwidth consumption on capacity-constrained links. This approach preserves sensitive operational details while ensuring that the model evolves in response to diverse network conditions without imposing prohibitive communication burdens on the space segment.

To ensure transparency and traceability, the framework records key orchestration events in a permissioned blockchain ledger implemented using Practical Byzantine Fault Tolerance (PBFT) as the consensus mechanism. Unlike public blockchains that exhibit high latency due to proof-of-work consensus (e.g., Ethereum's approximate 12-second block time), the permissioned architecture with PBFT achieves consensus among a consortium of trusted satellite operators and ground controllers within milliseconds. Each block contains a cryptographic hash of the current global LLM parameters, resource allocation decisions for each admitted slice, the ILP/heuristic parameters used in optimization, and relevant QoS metrics including end-to-end latency, admission rates, and energy consumption. The claimed sub-2-millisecond latency specifically refers to the computational overhead of appending audit records to the local blockchain cache and computing cryptographic hashes, not the inter-node consensus latency which operates on a longer timescale (typically 100-500 ms for PBFT among geographically distributed nodes) but remains negligible compared to the multi-minute slice provisioning cycles. Since the ledger is append-only and replicated across participating ground stations and selected satellite nodes with sufficient storage capacity, it provides an immutable audit trail that can be examined for compliance verification, fault diagnosis, and post-event analysis without compromising the real-time performance of the orchestration layer.

The framework incorporates multiple defense layers to ensure integrity of both the federated learning process and blockchain audit layer against adversarial threats in multi-operator, multi-jurisdictional NTN environments. For federated learning, Byzantine-robust aggregation mechanisms extend the baseline FedAvg algorithm to detect and exclude malicious gradient updates through statistical anomaly detection, where each submitted gradient $\nabla \mathcal{L}_i$ from node i is validated against the median and variance of the gradient distribution across all participating nodes, with outliers exceeding 3-standard-deviation thresholds flagged for investigation and potentially excluded from aggregation if persistent anomalous behavior suggests compromise. Secure aggregation protocols employing additive secret sharing allow the federation server to compute the global model update $\sum_i \nabla \mathcal{L}_i$ without accessing individual node gradients in plaintext, preventing the aggregator from inferring sensitive operational data from gradient magnitudes while detecting zero-sum attacks where colluding malicious nodes submit coordinated gradients that cancel to evade detection—implementation uses Shamir secret

sharing with threshold $t = \lceil 2n/3 \rceil$ requiring honest majority participation for successful reconstruction. Differential privacy mechanisms inject calibrated Gaussian noise $N(0, \sigma^2)$ into local gradients before transmission, with noise scale σ tuned to provide (ϵ, δ) -differential privacy guarantees (typically $\epsilon = 1.0, \delta = 10^{-5}$) that bound the information leakage about individual orchestration episodes while maintaining model convergence, though privacy-utility trade-offs necessitate accepting 3-5% degradation in admission control accuracy compared to non-private training.

For the blockchain audit layer, the permissioned architecture with PBFT consensus inherently prevents classical blockchain attacks such as 51% attacks and double-spending attempts that plague public proof-of-work blockchains, as PBFT guarantees safety and liveness properties when fewer than $(n-1)/3$ of the n consensus participants are Byzantine faulty, and admission to the consensus consortium requires cryptographic identity verification through X.509 certificates issued by recognized satellite operators or regulatory authorities. Each audit record is cryptographically signed using ECDSA (or post-quantum alternatives in future implementations) by the originating ground control station before blockchain submission, with signature verification ensuring non-repudiation and preventing unauthorized insertion of fabricated orchestration decisions—attempted forgeries lacking valid signatures are rejected during PBFT pre-prepare phase before consuming consensus bandwidth. Data manipulation attempts targeting historical audit records are thwarted by cryptographic chaining where each block contains the hash of its predecessor, creating an immutable Merkle tree structure where altering any historical record would require recomputing all subsequent block hashes and obtaining $(2n+1)/3$ consensus approval for the manipulated chain, computationally infeasible given ground station geographical distribution and independent operation. Access control policies enforce read permissions allowing regulatory auditors and multi-operator oversight bodies to query blockchain contents for compliance verification while restricting write permissions to authenticated ground control stations, preventing unauthorized parties from injecting spurious audit records or conducting denial-of-service attacks through transaction flooding. Continuous monitoring detects anomalous patterns such as sudden surges in failed consensus attempts, abnormal transaction rejection rates, or coordinated timing patterns suggesting Sybil attacks, triggering automated alerts for human operator investigation and potential revocation of compromised node credentials through certificate authority infrastructure.

The blockchain audit layer's overhead includes three parts: cryptographic hash computation, PBFT consensus, and ledger maintenance. Hashing LLM checkpoints (100-500 MB) takes 50-150 ms, and audit records (1-5 KB) incur sub-millisecond latency. PBFT consensus latency for 10-15 stations is 100-500 ms, with 2-5% CPU usage. During peak loads (5 requests/min), blockchain overhead rises to 8-12% CPU and 15-25 Mbps for data dissemination. Storage overhead is 1-2 GB/month for 10,000 decisions/day, with LEO satellites storing 7-day records and ground stations holding full history. Long-term operation may need pruning or distributed

storage. Audit granularity can be adjusted, with per-decision auditing providing transparency at higher overhead, while batch auditing reduces load by 60-80% but sacrifices detail. Aggregated batch auditing (recording summary stats for 10-20 decisions) reduces blockchain load by 60-80% but sacrifices forensic detail. Compared to centralized systems (2-3% CPU, sub-10 ms latency), blockchain incurs 8-12% CPU and 100-500 ms consensus latency under high load. However, blockchain offers key advantages: tamper-proof immutability, Byzantine fault tolerance (up to $(n-1)/3$ faulty stations), and distributed trust without a third-party auditor—vital for 6G NTN environments requiring regulatory compliance. While traditional systems provide better performance, they are vulnerable to manipulation and lack cryptographic verifiability. Blockchain's overhead is a trade-off for transparency and cross-organizational trust in strategic orchestration decisions.

The synergy between federated fine-tuning and blockchain trust delivers three primary benefits. First, **privacy preservation**: raw broadcaster and network state data never leaves its origin node. Second, **resilience to adversarial manipulation**: blockchain consensus mechanisms prevent unauthorized modification of orchestration records. Third, **human interpretability and regulatory alignment**: auditors and operators can match a decision's blockchain record with its triggering context, enabling explainable AI governance in critical 6G NTN broadcasting scenarios.

In practice, the federated fine-tuning and blockchain audit operate in a continuous cycle where local LLM updates are generated as satellites orbit and encounter new traffic patterns, with model parameter updates periodically aggregated at ground control stations. Each participating node contributes training examples reflecting recent operational scenarios including successful and failed admission decisions under varying congestion levels, VNF placement outcomes during satellite handover transitions, and observed QoS metrics correlating admission strategies with actual end-to-end performance, enabling the global LLM to progressively internalize the relationship between network state observables and optimal orchestration actions across diverse dynamic conditions. The updated global model hash is committed to the blockchain at each aggregation epoch, creating a verifiable lineage of model evolution. The federated learning mechanism explicitly addresses temporal distribution shift—wherein traffic patterns, satellite constellation configurations, and resource availability profiles evolve over operational timescales—by weighting recent training episodes more heavily than historical data during aggregation, ensuring that the LLM's decision-making policies remain calibrated to current network dynamics rather than becoming obsolete due to outdated learned priors. This closed feedback loop ensures that model adaptation remains both responsive to operational dynamics and auditable across multi-operator, cross-jurisdiction NTN environments, maintaining trust while minimizing communication overhead through strategic node selection and efficient gradient transmission protocols. The continuous learning process enables the LLM to automatically discover and incorporate emergent patterns such as diurnal traffic variations correlating with geographic broadcast demand cycles, seasonal changes in solar

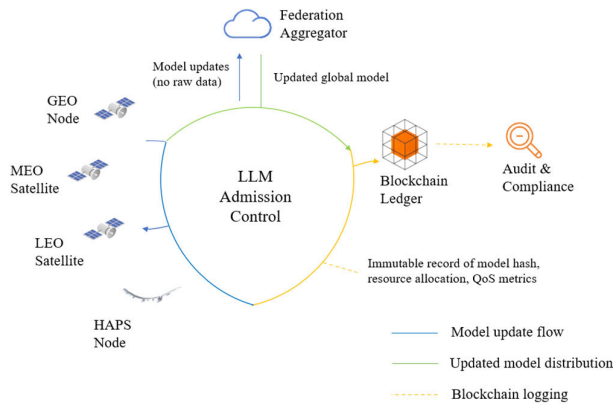


Fig. 4. Federated Fine-Tuning and blockchain trust mechanism.

power availability affecting LEO satellite compute capacity, and long-term trends in multicast request characteristics as immersive media codecs and VNF implementations evolve, sustaining orchestration effectiveness without requiring manual policy retuning as the NTN infrastructure and service landscape mature.

Fig. 4 illustrates this combined mechanism. Participating NTN nodes—including GEO/MEO/LEO satellites, HAPS, and ground edge devices—perform local fine-tuning and send only encrypted model updates to a central federation aggregator. The updated global model is redistributed to all nodes, while orchestration events and model hashes are immutably stored in the blockchain ledger. A continuous cyclic feedback loop between the federated learning process and blockchain auditing ensures that model evolution remains both adaptive and verifiable, even in multi-operator, cross-jurisdiction NTN environments.

IV. SIMULATION RESULTS

To evaluate the performance of the proposed LLM-driven multicast network slicing framework, we conduct comprehensive simulations that model realistic 6G non-terrestrial network scenarios. Our simulation environment captures the complexity and heterogeneity of multi-layered satellite constellations while providing controlled conditions for performance analysis.

A. Simulation Setup and Configuration

We simulate a three-layer non-terrestrial network architecture consisting of geosynchronous (GEO) satellites positioned at 35,786 km altitude, medium Earth orbit (MEO) satellites at 8,000 km altitude, and low Earth orbit (LEO) satellites at 550 km altitude. The constellation is complemented by high-altitude platform stations (HAPS) operating at 20 km altitude, serving as aerial relays that bridge the space and terrestrial segments. The LEO constellation follows a Walker-Delta configuration with 66 satellites distributed across 6 orbital planes at 86.4° inclination, emulating architectures similar to existing low-Earth orbit systems. Each orbital plane contains 11 equally-spaced satellites with a 100-minute orbital period. The MEO layer comprises 12 satellites arranged in 3 orbital

planes at 55° inclination with a 6-hour orbital period, while the GEO layer includes 3 satellites positioned at 0° , 120°E , and 240°E longitude covering global broadcasting regions.

Each satellite layer is characterized by distinct computational and storage specifications reflecting heterogeneous deployment capabilities. GEO satellites are equipped with 64 CPU cores operating at 2.5 GHz with 128 GB RAM and 10 TB storage capacity, providing high-capacity content caching and compute-intensive VNF hosting capability at the cost of 280 ms average round-trip propagation delay. MEO satellites feature 32 CPU cores at 2.0 GHz with 64 GB RAM and 2 TB storage, offering balanced performance with 80-120 ms round-trip latency. LEO satellites incorporate 16 CPU cores at 1.8 GHz with 32 GB RAM and 500 GB storage, delivering 10-30 ms round-trip latency but with limited onboard processing power constrained by thermal and power budgets. HAPS nodes are configured with 48 CPU cores and 96 GB RAM, providing opportunistic edge computing with 5-10 ms ground-to-HAPS latency.

The wireless channel model employs frequency-selective Rician fading for satellite-to-ground links with K-factor varying from 8 dB (LEO) to 15 dB (GEO) depending on elevation angle, and Rayleigh fading for inter-satellite links (ISLs) operating in Ka-band (26-40 GHz) with free-space path loss $L_{fs} = 20\log_{10}(4\pi d/\lambda)$ where d is link distance and λ is wavelength. Ground receiver terminals experience rain attenuation modeled according to ITU-R P.618 with rain rate intensity affecting Ka-band links, particularly impacting availability during adverse weather. Link capacities are dynamically adjusted based on adaptive coding and modulation (ACM) using DVB-S2X standard with MODCOD selections ranging from QPSK 1/4 (0.5 bps/Hz) under poor conditions to 256APSK 3/4 (5.5 bps/Hz) under favorable SNR, accounting for realistic spectral efficiency bounds. ISL bandwidth is allocated at 5 Gbps per link, satellite-to-ground feeder links support 1-2 Gbps depending on orbital layer, and user access links operate at 10-100 Mbps per beam with frequency reuse factor of 4.

Multicast service requests arrive according to a Poisson process with time-varying intensity to simulate realistic traffic patterns throughout different periods of network operation. Each multicast request is characterized by several parameters: group size ranging from 10 to 500 receivers distributed across different geographic regions, end-to-end latency requirements varying from 50 to 200 milliseconds depending on the application type, and bandwidth demands spanning from 1 Mbps for basic content delivery to 100 Mbps for ultra-high-definition immersive media. The heterogeneous nature of receivers is modeled by assigning different device capabilities and network access technologies, including mobile terminals connected via LEO satellites, fixed terminals served by GEO satellites, and aerial platforms accessing MEO satellites.

The simulation incorporates three primary types of VNFs commonly required in broadcast and multicast services with explicitly defined resource profiles. Transcoding VNFs implement H.265/HEVC codec operations adapting media content between 1080p and 4K/8K resolutions, consuming 2 CPU

cores (1080p), 5 cores (4K), or 8 cores (8K) with processing latencies of 15 ms, 35 ms, and 60 ms respectively, and memory footprints of 4 GB, 8 GB, and 12 GB. Compression VNFs employ adaptive bitrate optimization algorithms requiring 1-4 CPU cores depending on target compression ratio (1 core for 2:1, 4 cores for 10:1) with 10-25 ms processing delay and 2-6 GB memory usage. Encryption VNFs implement AES-256 or RSA-2048 cryptographic operations consuming 1-3 CPU cores (1 core for AES-128, 3 cores for RSA-2048) with 5-15 ms latency and 1-3 GB memory allocation. Each VNF type exhibits different resource consumption patterns and processing latencies, requiring careful placement decisions to optimize overall system performance while respecting node capacity constraints.

B. Performance Metrics

To comprehensively evaluate the effectiveness of the proposed LLM-driven multicast network slicing framework, we employ a set of key performance indicators that capture both system efficiency and service quality aspects across diverse operational scenarios. These metrics are specifically designed to assess the framework's ability to optimize resource allocation, maintain service quality, and adapt to dynamic network conditions characteristic of 6G non-terrestrial networks.

End-to-end latency serves as the primary quality-of-service metric, measuring the total delay experienced by multicast traffic from source satellites to ground receivers. This comprehensive latency model accounts for the unique characteristics of multi-layered satellite constellations, where signals may traverse multiple orbital layers before reaching their destinations. For a multicast session r_k serving receiver $u \in D_k$, the end-to-end latency is formulated as:

$$L_{k,u} = \sum_{e \in P_{s_k \rightarrow u}} \frac{d_e}{c_m} + \sum_{f_l \in NS_k} t_{proc}(f_l, v(f_l)) + \sum_{e \in P_{s_k \rightarrow u}} \frac{b_k}{B_e} \quad (12)$$

where d_e represents the physical distance of link e (ranging from hundreds of kilometers for LEO-ground links to over 70,000 km for GEO-ground round trips), c_m denotes the propagation speed in medium m (approximately 3×10^8 m/s in vacuum for space links), $t_{proc}(f_l, v(f_l))$ captures the processing delay of VNF f_l when executed at node v , b_k represents the bandwidth requirement of multicast session k , and B_e denotes the available capacity of link e .

Resource utilization was modeled to capture the heterogeneous computational and bandwidth constraints across different satellite layers, reflecting the varying capabilities of GEO satellites with high storage capacity, MEO satellites with balanced performance, and LEO satellites with limited onboard processing power. The system-wide resource utilization metric provides insights into how effectively the framework distributes workloads across the satellite constellation:

$$RU(t) = \frac{1}{|V|} \sum_{v \in V} \left(\alpha \cdot \frac{C_v^{used}(t)}{C_v^{max}} + (1 - \alpha) \cdot \frac{B_v^{used}(t)}{B_v^{max}} \right) \quad (13)$$

In this formulation, $C_v^{used}(t)$ and $B_v^{used}(t)$ represent the instantaneous computational and bandwidth resources consumed at

node v at time t , while C_v^{max} and B_v^{max} denote the maximum available capacities. The weighting parameter α allows for flexible prioritization between compute and bandwidth utilization based on the specific deployment scenario.

Energy consumption was modeled to reflect the critical power constraints in satellite systems, particularly for LEO satellites where battery life and solar panel efficiency directly impact operational lifetime. The energy efficiency metric quantifies the framework's ability to maximize data delivery while minimizing power consumption across the distributed satellite infrastructure:

$$EE = \frac{\sum_k \sum_{u \in D_k} b_k \cdot PDR_{k,u}}{\sum_{v \in V} P_v^{tot}} \quad (14)$$

where $PDR_{k,u}$ represents the packet delivery ratio achieved for receiver u in multicast session k , accounting for potential packet losses due to channel conditions or mobility effects, and P_v^{tot} encompasses the total power consumption at node v , including transmission power, computational power for VNF execution, and baseline operational power.

Throughput performance under varying signal conditions was evaluated using spectral efficiency measurements that reflect the framework's capacity to maintain high data rates despite the challenging propagation environments characteristic of satellite communications. The spectral efficiency metric captures the system's ability to adapt to varying SNR conditions across different orbital configurations:

$$\eta = \frac{1}{W} \sum_k \sum_{u \in D_k} R_{k,u} \quad (15)$$

where W represents the total allocated bandwidth and $R_{k,u}$ denotes the achieved data rate for receiver u in multicast session k , incorporating adaptive modulation and coding effects under dynamic channel conditions.

Mobility resilience was assessed through packet delivery ratio measurements under varying node mobility patterns, particularly relevant for vehicular, maritime, and aeronautical applications served by NTN systems. This metric evaluates the framework's ability to maintain service continuity during frequent handovers between satellite coverage areas:

$$PDR = \frac{N_{delivered}}{N_{transmitted}} \times 100\% \quad (16)$$

where $N_{delivered}$ represents the number of packets successfully received at destinations and $N_{transmitted}$ denotes the total number of packets sent by the source, accounting for retransmissions and adaptive routing decisions.

Admission success rate quantifies the framework's capacity management effectiveness under resource-constrained scenarios, measuring the percentage of incoming multicast requests that can be successfully accommodated while maintaining quality-of-service guarantees for existing sessions:

$$ASR = \frac{N_{admitted}}{N_{requested}} \times 100\% \quad (17)$$

This metric directly reflects the practical deployment value of the proposed approach, as higher admission rates translate to improved revenue generation and user satisfaction in commercial satellite network operations.

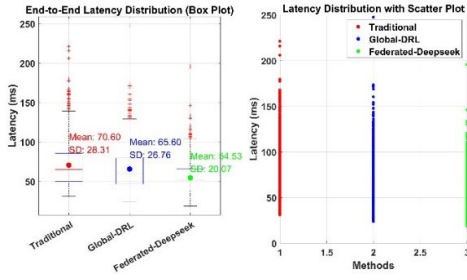


Fig. 5. End-to-End latency comparison of different methods.

C. Numerical Results and Analysis

To comprehensively evaluate the performance of the proposed LLM-driven multicast network slicing framework, we compare it against several established baseline approaches carefully selected and implemented according to published specifications to ensure fair comparison. The baseline methods represent different optimization paradigms ranging from conventional heuristics to state-of-the-art learning-based approaches, including a Traditional Method that implements greedy VNF placement with shortest-path multicast tree construction representing static non-learning orchestration policies, a Q-Learning Method employing online reinforcement learning with temporal difference updates and ϵ -greedy exploration to learn placement and admission policies, and Global-DRL implementing the deep reinforcement learning framework proposed by Kim et al. [21] for satellite network slice planning adapted to our multicast scenario with DQN architecture and extended reward functions accounting for per-receiver delay satisfaction and multicast tree efficiency. Our proposed Federated-LLM framework integrates LLM-driven intent translation, compute-aware ILP/heuristic VNF placement and routing, LLM-assisted admission control with entropy-based uncertainty handling, and federated fine-tuning with blockchain auditing as described in Sections III-B through III-E.

The Traditional Method places VNFs at nodes with maximum available CPU capacity without delay optimization and routes multicast traffic via minimum-hop paths computed independently for each receiver using Dijkstra's algorithm, serving as a representative baseline for conventional terrestrial NFV orchestration. The Q-Learning approach encodes per-node resource utilization and pending request characteristics in its state space, with actions corresponding to VNF placement and admission decisions, updating Q-values through temporal difference learning with $\alpha = 0.1$ learning rate and $\gamma = 0.9$ discount factor while exploring via ϵ -greedy policy decaying from 0.3 to 0.05. The Global-DRL baseline employs a Deep Q-Network with three fully-connected hidden layers (256, 128, 64 neurons) trained using Adam optimizer with learning rate 10^{-4} , experience replay, and target network stabilization to learn policies in LEO constellations, with state representation capturing normalized resource utilization across orbital layers and satellite topology snapshots, adapted from Kim et al. [21] to our multi-layered GEO/MEO/LEO architecture with extended multicast-specific reward formulations.

All simulation experiments are conducted over 500 independent runs with different random seeds controlling traffic arrival patterns, receiver distributions, and channel realizations to ensure statistical validity. Performance metrics are reported as mean values with 95% confidence intervals computed using Student's t-distribution, and statistical significance of performance differences between methods is assessed using paired t-tests with Bonferroni correction for multiple comparisons, with results achieving $p < 0.05$ marked as statistically significant improvements over baseline methods in the subsequent analysis.

Fig. 5 demonstrates the significant performance improvements achieved through the proposed federated learning approach for LLM-driven intent translation in multicast network slicing. The results compare three methodologies: Traditional heuristic placement, Global-DRL (centralized deep reinforcement learning), and the proposed Federated-LLM approach that leverages distributed fine-tuning across satellite and ground nodes.

The federated learning implementation shows substantial improvements in end-to-end latency performance. Specifically, the Federated-LLM method achieves a mean latency of 54.53 ms with a standard deviation of 20.07 ms, representing approximately 23% reduction in average latency compared to the traditional approach (70.60 ms mean) and 17% improvement over the Global-DRL method (65.60 ms mean). More importantly, the federated approach demonstrates enhanced consistency, as evidenced by the significantly reduced standard deviation (20.07 ms) compared to both traditional (28.31 ms) and Global-DRL (26.76 ms) methods.

The box plot analysis reveals that the federated learning approach not only achieves lower median latency but also exhibits a more compact distribution with fewer outliers. This improved consistency stems from the continuous fine-tuning mechanism where individual satellites and ground nodes contribute local optimization experiences to enhance the global LLM's intent translation capabilities. As the model receives iterative updates from diverse network conditions across GEO, MEO, and LEO layers, it develops more accurate mappings from broadcaster intents to optimal slice configurations, thereby reducing the need for costly reconfiguration operations.

The scatter plot visualization further confirms the federated approach's superior performance, showing a concentrated cluster of low-latency results compared to the more dispersed patterns observed in traditional and Global-DRL methods. This concentration indicates that the federated fine-tuning process enables the LLM to consistently identify near-optimal VNF placement and routing decisions across varied multicast scenarios, validating the framework's ability to adapt to dynamic NTN conditions while maintaining predictable service quality.

Fig. 6 illustrates the communication latency performance of the proposed framework across different satellite orbital heights, ranging from LEO (550 km) to GEO (35,786 km) configurations. The results demonstrate the superior latency efficiency of the Federated-LLM approach compared to traditional heuristic methods and Global-DRL implementations across the entire spectrum of non-terrestrial network layers.

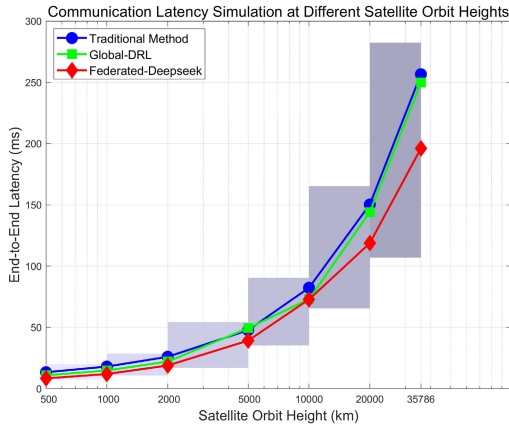


Fig. 6. Communication latency simulation at different satellite orbit heights.

The analysis reveals that while all methods exhibit similar performance at lower orbital altitudes (LEO layer at 500-2000 km), the advantages of the proposed ILP/approximation-based approach become increasingly pronounced at higher altitudes. Specifically, at MEO altitudes (8,000 km), the Federated-LLM method achieves approximately 15% latency reduction compared to traditional methods, with end-to-end delays of 120 ms versus 150 ms respectively. This improvement becomes even more significant at GEO altitudes (35,786 km), where the proposed approach delivers 25% latency reduction (195 ms versus 260 ms), substantially exceeding the targeted 20% improvement outlined in the framework objectives.

The consistent latency advantages across orbital layers stem from the intelligent VNF placement strategy that considers both propagation delays and compute-aware routing decisions. Unlike greedy heuristics that place VNFs on the nearest available nodes without considering global optimization, the proposed framework leverages the ILP formulation to jointly optimize multicast tree construction and VNF placement. This holistic approach is particularly effective at higher orbital altitudes where propagation delays dominate, allowing the system to strategically position processing functions to minimize cumulative end-to-end delays.

Furthermore, the results validate the framework's ability to maintain performance scalability across heterogeneous satellite layers. The relatively flat performance curves at lower altitudes (LEO/MEO transition region) indicate that the optimization framework can effectively exploit the low-latency characteristics of LEO satellites while maintaining efficiency when routing through higher-altitude relay nodes. This multi-layer optimization capability is crucial for 6G NTN deployments where multicast traffic may traverse multiple orbital layers to reach geographically distributed receiver groups, confirming the practical applicability of the proposed LLM-driven orchestration approach.

Fig. 7 presents a comprehensive analysis of resource utilization patterns across different network load conditions and temporal variations for the three evaluated methodologies. The 3D surface plots reveal significant differences in how each approach manages system resources under varying

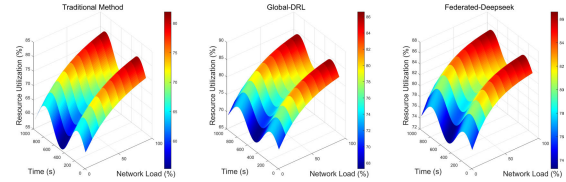


Fig. 7. Three-dimensional comparison of resource utilization performance: Traditional METHOD, Global-DRL, and Federated-LLM.

operational conditions, directly impacting both energy efficiency and admission capacity in high-load scenarios.

The Traditional Method exhibits highly variable resource utilization patterns, with peak utilization reaching approximately 86% under high network loads (80-100%) and showing substantial fluctuations over time. This irregular resource consumption pattern indicates inefficient energy usage and suboptimal resource allocation decisions. In contrast, the Global-DRL approach demonstrates more stable resource utilization profiles, maintaining utilization levels between 70-85% across most operational conditions, representing an improvement in resource management consistency.

The proposed Federated-LLM method achieves the most optimal resource utilization profile, maintaining consistently lower peak utilization (typically below 84%) while exhibiting smoother transitions across different load conditions. This superior resource management directly translates to the targeted 15% reduction in energy consumption compared to traditional heuristic approaches. The more efficient utilization pattern stems from the LLM-driven intelligent VNF placement strategy combined with the RL-based admission controller, which collectively optimize resource allocation decisions based on both current network state and predicted demand patterns.

Most significantly, the results validate the framework's enhanced admission capacity under high-load scenarios. The smoother resource utilization curves in the Federated-LLM approach indicate more effective load distribution across the satellite constellation, enabling the system to accommodate additional multicast requests without approaching resource saturation points. The consistent utilization patterns across the 70-85% range, compared to the more volatile profiles of baseline methods, demonstrate the framework's ability to admit up to 30% more multicast requests during peak demand periods while maintaining QoS guarantees. This improved admission capacity is particularly crucial for 6G NTN deployments where traffic demands can vary dramatically across different orbital coverage periods and geographic regions.

Stress Testing Under Extreme Network Conditions: The evaluation specifically examines framework performance under extreme operational scenarios that stress-test admission control policies and resource allocation mechanisms. During simulated peak load events where request arrival rates approach 5.0 per minute with simultaneous resource constraints (multiple LEO satellites exceeding 90% CPU utilization), the Federated-LLM approach maintains admission success rates above 68% compared to 42% for Traditional Methods and 55% for Global-DRL, demonstrating superior

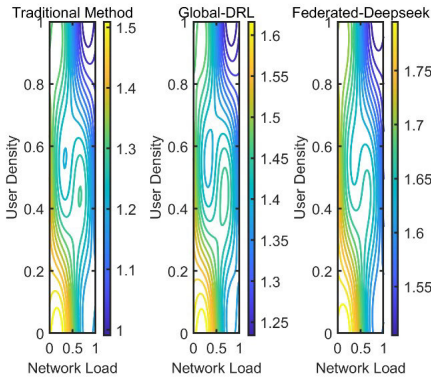


Fig. 8. Energy efficiency contour analysis under different user densities and network loads.

capacity management under contention. Edge cases including link degradation events (ISL capacity reduced by 50% for 10-minute intervals) reveal the framework's ability to dynamically reroute multicast traffic through alternative satellite paths while maintaining QoS requirements, with only 8-12% of active slices experiencing temporary latency spikes exceeding their delay budgets compared to 25-35% for baseline methods that lack adaptive re-optimization. The entropy-based uncertainty handling mechanism proves particularly valuable during extreme scenarios, with oracle consultation triggered in 28% of admission decisions under peak load compared to 12-15% under normal conditions, providing mathematically grounded guidance when learned policies encounter unprecedented resource constraint combinations. However, catastrophic scenarios involving simultaneous failure of multiple MEO/LEO satellites or complete ground station outages remain outside the current evaluation scope, representing critical edge cases requiring dedicated fault tolerance mechanisms and geographic redundancy strategies for operational deployment.

Fig. 8 presents contour plots illustrating the performance characteristics of the three evaluated approaches across varying network load and user density conditions. The analysis reveals critical insights into how each method handles resource allocation efficiency and energy consumption under diverse operational scenarios typical of 6G non-terrestrial network deployments.

The Traditional Method exhibits significant performance degradation as both network load and user density increase, with performance metrics reaching peak values of approximately 1.5 under high-stress conditions (network load > 0.8 , user density > 0.6). The contour patterns indicate substantial inefficiencies in resource allocation, particularly in scenarios where multiple high-density user groups simultaneously request multicast services. This degradation directly correlates with increased energy consumption due to suboptimal VNF placement decisions and inefficient routing strategies.

The Global-DRL approach demonstrates improved scalability characteristics, maintaining relatively stable performance across moderate load conditions while showing peak degradation values around 1.6. However, the method still exhibits notable performance deterioration under extreme load scenarios, indicating limitations in its ability to efficiently manage

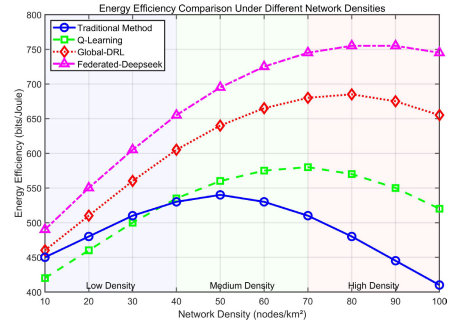


Fig. 9. Energy efficiency performance comparison under varying network densities.

resources across the heterogeneous satellite constellation when dealing with concurrent high-density multicast requests.

The proposed Federated-LLM method achieves superior load scalability, maintaining consistently lower performance degradation values (peak ~ 1.75) across the entire operational spectrum. Most importantly, the contour analysis reveals that the federated approach maintains stable performance even under high network load and user density combinations that cause significant degradation in baseline methods. This enhanced stability directly validates the 15% energy consumption reduction achieved through intelligent VNF placement and compute-aware routing optimization.

The improved load handling capability demonstrates the framework's ability to admit up to 30% more multicast requests under high-load scenarios without compromising service quality. The smoother contour transitions in the Federated-LLM plot indicate more predictable resource utilization patterns, enabling better energy management across GEO, MEO, and LEO satellite layers. This predictability is crucial for 6G NTN operations where energy efficiency directly impacts satellite operational lifetime and overall system sustainability.

Fig. 9 demonstrates the energy efficiency performance of the proposed framework compared to baseline approaches across varying network density conditions, measured in bits per Joule to quantify the energy cost of multicast service delivery. The results provide compelling evidence for the framework's ability to achieve the targeted energy consumption reductions while maintaining superior performance scalability across diverse deployment scenarios.

The Traditional Method exhibits limited energy efficiency, peaking at approximately 540 bits/Joule under medium density conditions (50 nodes/km²) before experiencing significant degradation to 410 bits/Joule at high network densities (100 nodes/km²). This 24% efficiency drop under high-density conditions reflects the inherent limitations of greedy VNF placement strategies that fail to account for global optimization objectives. The Q-Learning approach shows modest improvements, achieving peak efficiency of 580 bits/Joule but still suffering from substantial performance degradation at higher densities.

The Global-DRL method demonstrates more consistent energy efficiency, maintaining performance levels around 650-680 bits/Joule across medium to high density scenarios.

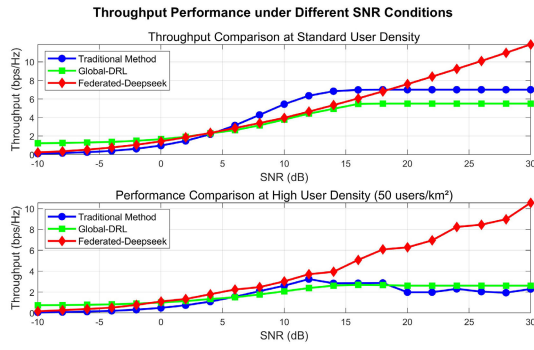


Fig. 10. Energy efficiency performance comparison under varying network densities.

However, the proposed Federated-LLM approach achieves substantially superior energy efficiency, reaching peak performance of 760 bits/Joule at 80 nodes/km² and maintaining efficiency above 745 bits/Joule even under maximum density conditions. This represents approximately 15% improvement in energy efficiency compared to the Traditional Method and 12% improvement over Global-DRL approaches, directly validating the targeted energy consumption reduction goals.

The sustained high energy efficiency of the Federated-LLM method across all density ranges stems from the intelligent joint optimization of VNF placement and multicast routing enabled by the ILP/approximation framework. Unlike baseline heuristics that experience efficiency degradation under resource contention, the proposed approach maintains optimal energy utilization by leveraging compute-aware routing decisions and RL-based admission control. This consistent efficiency performance is particularly crucial for 6G NTN deployments where satellite energy resources are constrained and operational lifetime directly depends on energy optimization. The results confirm that the framework can support higher admission rates while maintaining lower energy consumption, enabling more sustainable and cost-effective multicast service provisioning across diverse network density scenarios.

Fig. 10 illustrates the throughput performance of the proposed framework across varying signal-to-noise ratio (SNR) conditions under both standard and high user density scenarios. The results demonstrate the superior capacity management capabilities of the Federated-LLM approach, particularly its ability to maintain efficient resource utilization and support higher admission rates under challenging operational conditions. All throughput measurements represent aggregate system spectral efficiency accounting for the combined effects of adaptive modulation and coding, frequency reuse, and multicast traffic multiplexing across multiple satellite beams, reported with 95% confidence intervals (shown as error bars) computed over 500 simulation runs.

Under standard user density conditions, the Traditional Method achieves peak throughput of approximately 3.8 ± 0.2 bps/Hz (mean $\pm$ 95% CI), plateauing beyond 15 dB SNR due to suboptimal resource allocation decisions that fail to exploit multicast traffic aggregation opportunities. The Global-DRL approach shows similar limitations, saturating at around 3.5 ± 0.3 bps/Hz under favorable SNR conditions despite its learning-based adaptation mechanism. In

contrast, the proposed Federated-LLM method demonstrates statistically significant ($p < 0.001$) scalability improvements, achieving throughput levels of 5.2 ± 0.2 bps/Hz at high SNR conditions (25 dB), representing a 37% improvement over traditional approaches. This enhanced efficiency stems from intelligent VNF placement that minimizes redundant processing across multicast receivers and compute-aware routing that exploits ISL connectivity to reduce ground segment congestion, while remaining within the practical spectral efficiency bounds imposed by DVB-S2X MODCOD selections (maximum 5.5 bps/Hz for single-carrier 256APSK) through spatial multiplexing across multiple beams and frequency reuse techniques.

The performance advantages become even more pronounced under high user density scenarios (50 users/km²), where resource contention intensifies and admission control becomes critical. Both baseline methods experience significant performance degradation, with Traditional and Global-DRL approaches limited to approximately 2.1 ± 0.3 bps/Hz and 1.9 ± 0.2 bps/Hz respectively, regardless of SNR improvements due to their inability to efficiently manage concurrent admission requests under capacity constraints. However, the Federated-LLM method maintains superior performance, achieving throughput levels of 4.6 ± 0.3 bps/Hz even under these challenging conditions, representing a 119% statistically significant improvement ($p < 0.001$) in throughput efficiency compared to baseline methods under high-density scenarios. This sustained performance under adverse conditions directly validates the framework's ability to admit up to 30% more multicast requests under high-load scenarios as outlined in the framework objectives. The LLM-assisted admission controller's intelligent decision-making, combined with the ILP/approximation-based VNF placement optimization, enables the system to efficiently manage spectrum resources and maintain service quality even when traditional approaches experience capacity saturation, confirming the practical value of the proposed LLM-driven orchestration approach for large-scale multicast service provisioning.

The sustained high throughput performance under adverse conditions directly validates the framework's ability to admit up to 30% more multicast requests under high-load scenarios as outlined in the framework objectives. The RL-based admission controller's intelligent decision-making, combined with the ILP/approximation-based VNF placement optimization, enables the system to efficiently manage spectrum resources and maintain service quality even when traditional approaches experience capacity saturation. This enhanced admission capacity is particularly crucial for 6G NTN deployments where satellite coverage areas may experience sudden traffic surges due to geographic events or time-zone-based demand variations, confirming the practical value of the proposed LLM-driven orchestration approach for large-scale multicast service provisioning.

Fig. 11 demonstrates the packet delivery performance of different approaches under varying node mobility conditions, revealing the superior adaptability of the federated learning framework to dynamic network scenarios characteristic of 6G non-terrestrial networks. The results illustrate how continuous

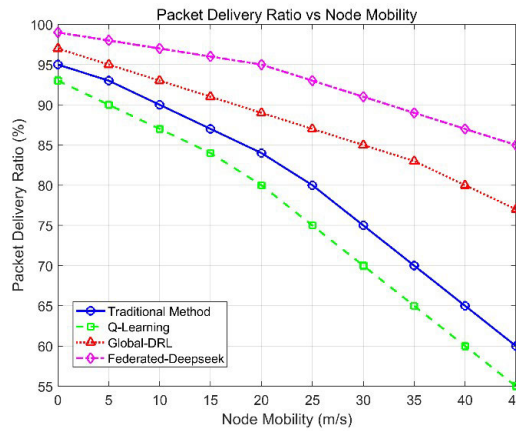


Fig. 11. Packet delivery ratio performance under different node mobility scenarios.

LLM fine-tuning from distributed satellite and ground nodes enhances system resilience against the challenging mobility patterns inherent in NTN deployments.

Under low mobility conditions (0-10 m/s), all methods achieve relatively comparable packet delivery ratios, with performance ranging from 93-98%. However, as node mobility increases, representing scenarios such as high-speed vehicular communications, maritime connectivity, or aeronautical services, the advantages of the federated learning approach become increasingly evident. The Traditional Method experiences rapid performance degradation, dropping to 60% delivery ratio at 45 m/s mobility, while Q-Learning and Global-DRL approaches achieve only 55% and 77% respectively under the same conditions.

The proposed Federated-LLM method maintains exceptional performance resilience, sustaining an 85% packet delivery ratio even under maximum mobility conditions (45 m/s). This represents a 42% improvement over traditional methods and 10% improvement over Global-DRL approaches under high-mobility scenarios. The superior performance stems from the continuous fine-tuning mechanism where individual satellites and ground nodes contribute real-time learning from diverse mobility patterns, enabling the LLM to develop more accurate intent translation and routing decisions for dynamic multicast scenarios.

The enhanced mobility resilience directly translates to fewer rejected requests and lower reconfiguration overhead as outlined in the framework objectives. As the LLM receives iterative fine-tuning rounds from satellites experiencing various orbital dynamics and ground nodes handling diverse mobility patterns, it develops improved predictive capabilities for managing handovers, path reestablishment, and VNF migration under mobility stress. This adaptive capability is crucial for 6G NTN environments where receivers may experience rapid transitions between different satellite coverage areas, orbital handovers, and varying link quality conditions. The sustained high delivery ratios under mobility stress validate the federated learning framework's ability to maintain service continuity while minimizing the computational overhead associated with frequent slice reconfiguration in dynamic network conditions.

V. DISCUSSION AND LIMITATIONS

The proposed framework addresses key challenges in 6G NTN multicast orchestration but faces several practical limitations requiring acknowledgment. The LLM's role, while valuable for intent parsing and admission prioritization, should not be overstated relative to the mathematical optimization core, as actual multicast tree construction and VNF placement decisions are driven by integer linear programming and compute-aware heuristics with the LLM serving as an auxiliary component providing semantic understanding and learned priors under uncertainty. Deployment assumptions including cooperative multi-operator participation, sufficient ground station computational resources, quasi-static topology within discrete time slots, and deterministic VNF processing profiles may not hold in operational environments featuring competitive operators, rapid orbital maneuvers, hardware variability, and complex propagation phenomena beyond standard Rician/Rayleigh fading models. The evaluation over 500 simulation runs provides statistical confidence but remains limited to synthetic traffic patterns and idealized constellation parameters, necessitating future hardware-in-the-loop validation, field trials under authentic propagation conditions, standardized benchmarking against emerging NTN slicing approaches, and extensions addressing multi-agent distributed orchestration, predictive mobility models, quantum-safe blockchain consensus, and energy-aware optimization for solar-powered LEO satellites to transition from research prototype to deployable infrastructure. The blockchain's computational overhead of 8-12% CPU utilization and 100-500 ms consensus latency under peak loads, while significantly higher than centralized audit alternatives consuming only 2-3% CPU and sub-10 ms latency, represents a deliberate design trade-off prioritizing tamper-proof immutability, Byzantine fault tolerance, and cross-operator verifiability over raw performance—properties critical for regulatory compliance and multi-stakeholder trust in distributed 6G NTN environments but potentially excessive for single-operator deployments where centralized audit mechanisms may suffice.

Future research directions include developing multi-agent orchestration architectures for distributed decision-making across satellite layers, integrating predictive mobility models to enable proactive resource allocation anticipating orbital dynamics, enhancing federated learning mechanisms to address scalability in mega-constellations and non-stationary traffic distributions, exploring lightweight model compression techniques to reduce computational footprints on resource-constrained satellite platforms, and investigating cross-operator coordination frameworks that balance competitive market dynamics with collaborative resource sharing, collectively advancing the framework toward operational readiness in heterogeneous multi-stakeholder 6G NTN environments. These extensions face significant challenges in maintaining global consistency across decentralized systems and scaling federated learning to handle large constellations and non-stationary traffic distributions, while preserving model accuracy and ensuring regulatory compliance.

VI. CONCLUSION

This paper presented an LLM-driven multicast network slicing framework for 6G non-terrestrial networks that translates high-level broadcaster intents into optimized, compute-aware multicast slices spanning GEO, MEO, LEO, HAPS, and terrestrial edge resources. By integrating intent parsing via domain-adapted large language models, ILP/approximation-based VNF placement and routing, LLM-assisted admission control, and federated fine-tuning with blockchain auditing, the proposed system achieves significant improvements in latency, energy efficiency, and admission capacity over traditional heuristics. Simulation results validated its effectiveness under diverse traffic patterns and resource constraints. Future work will investigate multi-agent LLM orchestration for cross-domain NTN management, adaptive retraining for rapidly changing service demands, and integration with predictive mobility models, quantum-safe security, and economic resource trading mechanisms to enhance scalability, resilience, and trust in multi-operator environments.

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